

Modality

Motivation[1]

Modalities in homotopy type theory are used to create and access subuniverses of a given type universe. These have significant applications throughout mathematics and computer science, and in particular can be used to create universes in which certain logical principles are true.

reflective subuniverse(反射 · 子宇宙)

- ▶ a subuniverse $U_{\circ}$ of U
- ▶ a reflector $\circ : U \rightarrow U_{\circ}$
- ▶ a unit $\eta : \prod_{X:U} X \rightarrow \circ X$

满足 **modal recursion**: for all $X : U$, $Y : U_{\circ}$, the map

$$- \circ \eta_X : (\circ X \rightarrow Y) \rightarrow (X \rightarrow Y)$$

is an equivalence. 这相当于, 对任何 $f : A \rightarrow B$, 其中 $B : U_{\circ}$, 存在唯一的 $\bar{f} : \circ A \rightarrow B$, 使得 $f = \bar{f} \circ \eta_A$

A commutative triangle diagram illustrating the modal recursion property. It consists of three nodes: A at the top-left, B at the top-right, and $\circ A$ at the bottom-left. A solid arrow labeled f points from A to B . A solid arrow labeled η_A points from A down to $\circ A$. A dashed arrow labeled $\exists! \bar{f}$ points from $\circ A$ up to B . The diagram shows that the function f can be factored through the reflection of its domain A via a unique function $\bar{f}$.

reflective subuniverse 的特点

Reflective subuniverses can be shown to have many stability properties. In particular, they are closed under Π , $+$, $\times$, and taking equality types. However, they do not quite constitute logical subuniverses as they are not necessarily stable under Σ .

modality

A **modality** is a reflective subuniverse (U_O, O, η) satisfying one of the following equivalent conditions:

1. U_O is closed under Σ .
2. O satisfies **modal induction**. That is for all $B : OA \rightarrow U_O$, the natural map

$$\prod_{x:OA} B(x) \rightarrow \prod_{x:A} B(\eta(x))$$

is an equivalence.

We call a type **modal** if the map η_x is an equivalence.

lex modality

A **lex modality** is a modality $\Box$ which preserves pullbacks.

Even modalities don't correspond to the external notion of subtopos. Lex modalities really do correspond to the external notion of subtopos and as such they behave well with respect to many of the structures in HoTT.

参考文献 I



Mark Damuni Williams. Projective Presentations of Lex Modalities.